Calidroid-Nav: Navigation & SMS Integration Documentation

Calidroid-Nav is an Android navigation app that receives GPS waypoints via SMS, processes them, and communicates directional commands to an Arduino via USB serial. It enables offline and satellite-assisted navigation for e-bikes, boats, or vehicles using simple text commands such as LEFT, RIGHT, or STRAIGHT.

Feature	Description
SMS Waypoint Parsing	Extracts latitude/longitude from Google Maps links received via SMS.
Turn Direction Logic	Determines turn direction based on phone heading vs. target bearing.
USB Serial Communication	Sends commands to Arduino: LEFT, RIGHT, STRAIGHT, HALT.
SMS Reply System	Replies to 'WHERE' SMS with current Google Maps location link.
Starlink/T-Mobile Fallback	Supports satellite text fallback for remote operation.

Setup Instructions
1. Install the Calidroid-Nav APK on your Android device.
2. Grant SMS permissions (RECEIVE_SMS, SEND_SMS).
3. Connect Arduino via USB OTG cable.
4. Verify serial communication using SimpleUsbTerminal.
5. Configure trusted phone numbers for waypoint SMS.
6. Test by sending a Google Maps link via SMS.

Command	Description	Example
WAYPOINT	Set navigation target via SMS link	https://maps.google.com/?q=33.9164,-11
WHERE	Request current location reply	Send 'WHERE' to get location link
LEFT / RIGHT / STRAIGHT	Sent via serial to Arduino for navigation turns	Arduino receives 'LEFT'

Arduino Integration
Arduino listens for serial messages from the Android device via USB.
Typical pin usage:
- Left motor PWM: GPIO 14
- Right motor PWM: GPIO 27
- Servo 1/2: GPIO 12/13
The microcontroller interprets the serial strings ('LEFT', 'RIGHT', etc.) and drives the motors accordingly.

Example SMS Flow:
1. User sends: *WAYPOINT https://maps.google.com/?q=33.9416,-118.4085*
2. App parses coordinates and compares to current GPS heading.
3. App sends 'RIGHT' or 'LEFT' via USB serial to Arduino.
4. If 'WHERE' is sent, app replies with current map link.